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Anti-staphylococcal potential of montanine-type derivatives from amaryllidaceae alkaloids alone and in combination with antibiotics

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Abstract

Aims **The rising wave of antimicrobial resistant organisms, together with the lack of novel group of antibacterial agents, represents a major concern in modern medicine. This study aims to investigate the anti-staphylococcal potential of two semi-synthetic derivatives of naturally occurring nitrogen-containing molecules [2-O-(3-methoxybenzoyl)-3-O-methylpancracine (1 nm) and 2-O-(3,5-dimethoxybenzoyl)-3-O-methylpancracine (2 nm)].**

Methods and results **The two semi-synthetic analogues were evaluated against reference bacterial strains and clinical bacterial isolates. Furthermore, the most active compound, 1 nm, was subjected to synergistic activity screening with the commercial antibiotics against MRSA. The toxicity profile of 1 nm was further explored using *in vitro* (HepG2 and HK-2 cells) and *in vivo* (*Galleria mellonella*) models, accompanied by an insight into its *in silico* physicochemical characteristics and cytotoxicity. Our results suggest that this derivative possesses promising activity against clinically relevant staphylococci, while showing a favourable safety and bioavailability profile. In addition, this compound showed synergistic or additive effects in combination with several clinically important antibiotics.**

Conclusions **1 nm demonstrates promising adjuvant anti-staphylococcal activity, with a favourable safety profile and an additive interaction with the last-resort antibiotic linezolid, supporting its potential to mitigate resistance development within combination strategies.**

Impact statement

With antimicrobial resistance continuing to outpace antibiotic discovery, effective antibiotic adjuvants represent a promising strategy to maximize the utility of existing treatments. This study identifies a novel anti-staphylococcal compound with a favourable safety profile and additive activity with linezolid, highlighting its translational potential for combination therapies aimed at improving treatment efficacy and helping preserve last-resort antibiotics.

Keywords multidrug-resistant bacteria, antibiotics, staphylococci, bacteria, antibiotic resistance

Introduction

Almost a century ago, the discovery of antibiotics significantly changed the rudiments of medicine and helped to control the most world-threatening illnesses. However, antimicrobial resistance (AMR), supported by the current overuse, misuse, and agricultural use of antimicrobial agents, brings infectious diseases back to global awareness (Muller 2022). Infections associated with AMR lead to prolonged hospital admissions, treatment failures, major increases in healthcare costs, and, sadly, a higher mortality rate. Since the effectiveness of antibiotics will continue to decline due to persistent trends of AMR, physicians might use last-resort groups of drugs such as carbapenems and polymyxins, which are expensive, have many different side effects, and are not broadly available (Dadgostar 2019). The need for new antibiotic agents' discovery is a long-lasting and inevitable problem worldwide.

Confirming this desperate urge, the World Health Organization (WHO) introduced the Global Action Plan on Antimicrobial Resistance in 2015 to increase awareness and understanding of AMR. Nowadays, the main focus lies in supporting research and

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developing new antimicrobial candidates or alternative strategies to combat AMR (Terreni *et al*. 2021).

Meanwhile, the antibiotic pipeline is gradually drying up, the spread of AMR is not slowing down. Clinicians are desperately challenged by the spread of superbugs, multidrug-resistant microorganisms, which are recognized as a serious global threat to human health. In 2024, the WHO published an updated list of pathogens, organized in three categories according to the urgency of the emergence (WHO 2024). Many of them are also gathered under the infamous acronym, ESKAPE—pathogens that have developed a great level of multi-resistance to agents from different antibiotic groups, such as oxazolidinones, lipopeptides, macrolides, fluoroquinolones, tetracyclines, β-lactams, β-lactam-β-lactamase inhibitor combinations, and even to the last line of defence, such as carbapenems, glycopeptides, and polymyxins (De Oliveira *et al*. 2020).

Despite the global effort to control the impact of staphylococcal infections upon the public and a healthcare environment enhanced by WHO's announcement in 2024, *Staphylococcus aureus* remains a threat to society especially in hospital environment, where it is transferred between patients, by contact with contaminated medical material and healthcare personnel or from exogenous environment and community (Santajit and Indrawattana 2016). While 30% of the population carries *S. aureus* as a harmless commensal, in specific conditions or immunocompromised patients, *S. aureus* represents a causative agent of wide-ranged diseases affecting most of the human tissues, such as skin, blood, bones, and organs—typically, heart and lungs. Such infections are complicated by prolonged hospital stays, higher mortality, and a large economic burden (Tong *et al*. 2015, Ajibola *et al*. 2023, Jiang *et al*. 2023).

Only shortly after discovery of methicillin as an alternative to treat penicillin-resistant strains, illustrious methicillin-resistant *Staphylococcus aureus* (MRSA) became the most frequently occurring resistant pathogen across continents. Furthermore, contrary to traditional variants, new clones of MRSA appear not only in hospital environments but also affect people without any predispositions (Lakhundi and Zhang 2018). This status and the increasing resistance level should lead to a deeper interest focused on new antimicrobials development.

No less of a problem is represented by coagulase-negative staphylococci (CoNS). Despite its benign reputation, this heterogeneous group is responsible for potentially critical infections in healthcare environments, predominantly in immunocompromised patients (Heilmann *et al*. 2019). The situation is, again, tragically worsened by a broad AMR of CoNS caused by various mechanisms of resistance, including biofilm formation. In addition, CoNS have been identified as a reservoir of resistance even against last-resort antibiotics, such as linezolid (LIN) or daptomycin (Marincola *et al*. 2021).

Invariable structures of synthetic anti-infective molecules and their short lifespan in combating bacterial resistance development have reversed the research trends in favour of natural products. Isolated natural compounds could either provide a source of anti-infective drugs or supply a potential core as 'lead molecules', which could be chemically modified while incorporating the active pharmacophores (Cushnie *et al*. 2014).

The observed antibacterial characteristics of natural product-derived molecules are assumed to be due to the evolution of secondary metabolite scaffolds through millennia, guaranteeing a competitive dominance over other microbes. Natural compounds likely evolved to infiltrate cell membranes and interact with particular receptors of microorganisms, with their chemically complex structure required to inhibit many bacterial protein targets. Thus, the discovery of well-optimized molecules with anti-infective properties from nature is not surprising (Butler and Buss 2006, Brown and Dawson 2015). Examples of introduced natural-derived antibacterial classes are β-lactam antibiotics such as penicillin, bacterial peptides such as bacitracin, aminoglycosides such as streptomycin, and macrolides such as erythromycin (Butler and Buss 2006).

Developing new anti-infective agents *via* chemical manipulation of naturally occurring scaffolds of antibiotics has gained popularity over the years. From 1995 to 2005, from 22 FDA (Food and Drug Administration) approved antibacterial drugs, 12 were natural product-derived drugs belonging to five antimicrobial classes (β-lactams, streptogramins, macrolides, tetracyclines, and lipopeptides). The remaining drugs were synthetic compounds, limited to only two antibacterial classes, with quinolones accounting for nine approvals (Butler and Buss 2006). In addition, between 2010 and 2022, of 14 small-molecule antibacterials approved by the FDA, nine were derivatives of naturally occurring scaffolds, highlighting the role of derivatization of natural frameworks in antibacterial drug discovery (Jung and Gademann 2023).

Alkaloids are one of the largest groups of secondary metabolites produced by plants, characterized by diverse nitrogen-containing chemical structures. Although they are famous as poisonous compounds, yet have been exploited in medicine in many cases (Shah and Seth 2010).

Given the various bioactivities assigned to alkaloids, researchers have persistently evaluated their antibacterial properties, leading to the introduction of promising antibacterial activity of alkaloids, particularly from the indole, indolizidine, piperazine, quinoline, isoquinoline, and quinolone classes, with minimum inhibitory concentration (MIC) values equal to or below 10 $\mu g\,ml^{-1}$ (Cushnie *et al*. 2014, Debnath *et al*. 2018). The isoquinoline group of alkaloids, despite the significant chemical variation exhibited, shares a common isoquinoline core and is predominantly distributed in *Papaveraceae, Berberidaceae, Ranunculaceae,* and *Amaryllidaceae* plants (Jin and Yao 2019, Yan *et al*. 2021).

The *Amaryllidaceae* family represents a plentiful source of isoquinoline core-bearing molecules, commonly referred to as *Amaryllidaceae* alkaloids (AAs) (Kornienko and Evidente 2008, Nair *et al*. 2017, Berkov et al. 2020). Allegedly, extensive utilization of this family in folk medicine is largely attributed to the presence of AAs regarding their verified biological values in neurodegenerative diseases, antitumour properties, and antimicrobial activity (Ding *et al*. 2017, Nair *et al*. 2017). To date, the majority of submicromolar antibacterial activity observed by AAs are predominantly reported from narciclasine-, lycorine-, and α-crinane-type, against a range of Gram-negative and Gram-positive pathogens, especially *Bacillus subtilis, Klebsiella pneumoniae*, *Escherichia coli*, *Helicobacter pylori, Pseudomonas aeruginosa*, *Staphylococcus epidermidis*, and *S. aureus* (Nair *et al*. 2017).

Montanine-type alkaloids are among the rare scaffolds of AAs and yet compelling as potential bioactive molecules (Govindaraju *et al*. 2018). However, the montanine-type nucleus has been proposed to be achievable easily through a semi-synthetic rearrangement of widely available haemanthamine (Fig. 1) (Inubushi *et al*. 1960, Cedrón *et al*. 2009, Govindaraju *et al*. 2018). This

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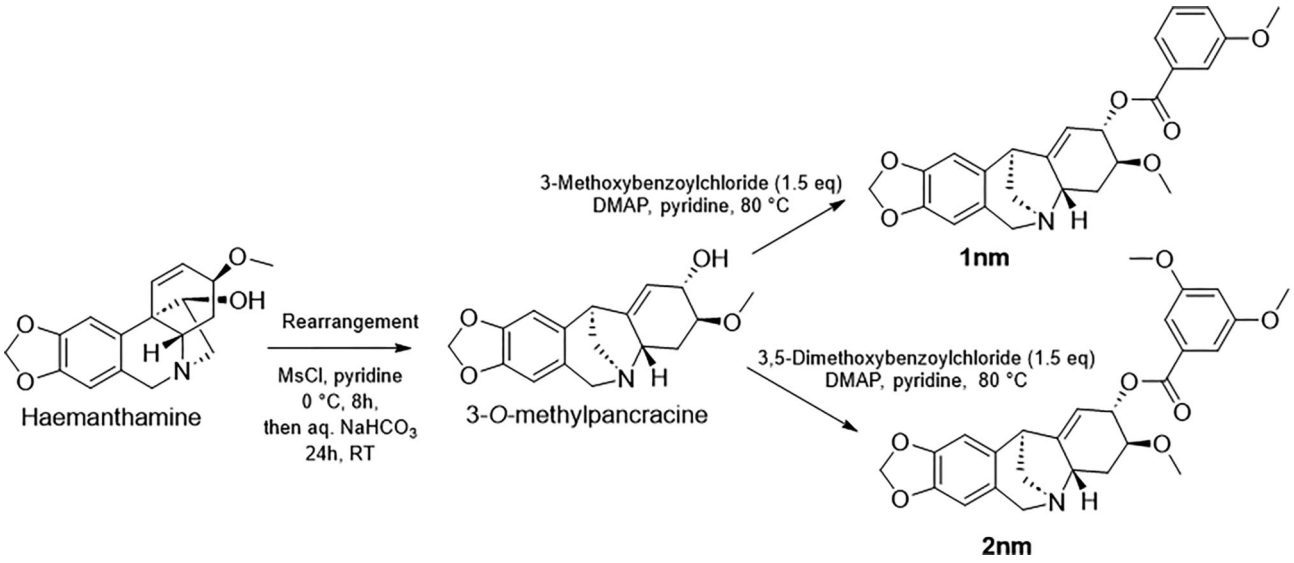


Figure 1 Reaction scheme for haemanthamine ring rearrangement to montanine-type AAs core and preparation of title compounds, 2-O-(3-methoxybenzoyl)-3-O-methylpancracine (1 nm) and 2-O-(3,5-dimethoxybenzoyl)-3-O-methylpancracine (2 nm).

approach has offered an accessible source of the montanine backbone (as in 3-*O*-methylpancracine), enabling further derivatization and allowing research groups to explore this intriguing scaffold (Govindaraju *et al*. 2018, Maafi *et al*. 2021a, 2021b). Interestingly, derivatization of montanine-type core has led to a significant enhancement of activity in this class against *Mycobacterium tuberculosis* H37Ra, *Mycolicibacterium aurum*, and *Mycolicibacterium smegmatis*. Similarly, the same outcome has been observed with other AAs such as galanthamine, maritidine, and vittatine (Maafi *et al*. 2021a).

Given the known bioactivity of montanine-type AAs and leveraging the intramolecular transformation of the haemanthamine core into the montanine-type scaffold, our research group has developed a library of semi-synthetic derivatives of these alkaloids and evaluated their biological properties (Maafi *et al*. 2021a, 2021b). Compounds 1 nm and 2 nm (Fig. 1) are selected hits from this library based on our positive assessment on the preliminary antibacterial screening. In the previous study, compounds 1 nm (previously published as '1f') were subjected to cholinesterase inhibitory screening to contribute to a comprehensive structure–activity relationship (SAR) analysis (Maafi *et al*. 2021b). Moreover, in this study, both compounds were also included in preliminary antimycobacterial screening, which exhibited satisfactory results ($MIC_{1nm} = 66.2$ μM, $MIC_{2nm} = 31.1$ μM, both for *M. tuberculosis* H37Ra). Pilot screening provided a rationale for extending the antimicrobial activity investigation of 1 nm and 2 nm against a broader spectrum of microorganisms. In this study, 1 nm and 2 nm are subjected to screening against reference strains, followed by bacteria isolated from clinical samples, with particular emphasis on pathogens known for their high level of resistance.

As an alternative to the conventional, long, and expensive development of a new leading drug, novel strategies for the therapy of infectious diseases are increasingly being explored, such as the use of combination therapy.

Antibiotic combinations are already a common strategy in the treatment of severely ill patient, predominantly through the use of two or more antibiotic molecules, each capable of acting as an independent therapeutic agent. Other strategies may also involve a combination of an existing antibiotic agent with an adjuvant, which, acting on its own, may show little to non-therapeutic effect but confers important advantages when used alongside the antibiotic. In fact, adjuvants can not only potentiate or enhance the antimicrobial activity of primary therapeutic agent but also minimize or even prevent AMR (Boyd *et al*. 2021).

The present study aimed to evaluate two semi-synthetic montanine-type derivatives, 1 nm and 2 nm, which belong to a class of compounds known to exhibit a broad spectrum of biological activities, including antibacterial effects. Within the framework of this study, we sought to characterize their antimicrobial and other relevant biological properties to assess their potential as promising candidate antibiotics or adjuvants. Furthermore, water solubility test for 1 nm and 2 nm was conducted *in vitro* and further physicochemical and ADME-related (Absorption, Distribution, Metabolism, and Excretion) parameters were evaluated *in silico* for both compounds to discuss their drug-likeness.

Materials and methods

General chemistry methods

All solvents were pretreated using standard procedures prior to use. Reagents and catalysts were obtained from Merck, Germany and used as received, without further purification. The NMR spectra were measured in $CDCl_3$ on a VNMR S500 (Varian) spectrometer operating at 500 MHz for ^1H and 126 MHz for ^{13}C at an ambient temperature. The chemical shifts were recorded as δ values in parts per million (ppm). The residual signal of $CDCl_3$ at δ 7.26 ppm was used as the reference for ^1H NMR spectra, while the central signal at δ 77.0 ppm served as the reference for proton-decoupled ^{13}C NMR spectra. The coupling constant (J) is given in Hz. ESI-HRMS was obtained with a Waters Synapt G2-Si hybrid mass analyser of a quadrupole-time-of-flight (Q-TOF) type, coupled to a Waters Acquity I-Class UHPLC system. Purity assessment of deriva-

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tives was performed using a Waters HPLC-PDA-MS system (Waters Corporation, MA, USA), confirming >95% purity for both compounds (PDA detection, UV–Vis, uncalibrated). Thin-layer chromatography was carried out on Merck precoated silica gel 60 F254 plates, and compounds on the plates were observed under UV light (254 and 366 nm) and visualized by Dragendorff's reagent spraying.

Preparation of 3-*O*-methylpancracine

3-*O*-methylpancracine was prepared as previously described, using ample quantities of haemanthamine isolated from *Narcissus pseudonarcissus* cv. Dutch Master bulbs (Hulcová *et al*. 2019, Maafi *et al*. 2021a, 2021b). Spectroscopic data of 3-*O*-methylpancracine corresponds with data reported before (Maafi *et al*. 2021b). The reaction was repeated multiple times to obtain sufficient quantities of 3-*O*-methylpancracine for the preparation of compounds 1 nm and 2 nm, if required.

Preparation of 1 nm and 2 nm

1 nm and 2 nm were prepared according to the previously described procedure (Kohelová *et al*. 2019, Al Shammari *et al*. 2021).

Physical, spectroscopic, and spectrometric data of derivatives 1 nm and 2 nm

2-*O*-(3-methoxybenzoyl)-3-*O*-methylpancracine (1 nm)

Spectrometric data of 1 nm corresponds to previously published records (Maafi *et al*. 2021b).

2-*O*-(3,5-dimethoxybenzoyl)-3-*O*-methylpancracine (2 nm)

Yield 96%; pale yellow amorphous solid; $[\alpha]_D^{28} = +27.16^\circ$ (*c* 0.27, $CHCl_3$); **^{1}H NMR** (500 MHz, $CDCl_3$) δ: 7.18 (d, $J = 2.3$ Hz, 2H), 6.66 (t, $J = 2.3$ Hz, 1H), 6.55 (s, 1H), 6.47 (s, 1H), 5.88 (d, overlap, $J = 13.4$ Hz, 1H), 5.88 (d, overlap, $J = 13.4$ Hz, 1H), 5.58–5.53 (m, 1H), 5.40–5.35 (m, 1H), 4.36 (d, $J = 16.7$ Hz, 1H), 3.87 (d, overlap, $J = 16.7$ Hz, 1H), 3.84 (s, overlap, 6H), 3.70–3.65 (m, 1H), 3.48–3.40 (m, overlap, 1H), 3.46 (s, overlap, 3H), 3.31 (d, $J = 2.4$ Hz, 1H), 3.12 (dd, $J = 11.2$, $J = 2.4$ Hz, 1H), 3.07 (d, $J = 11.2$ Hz, 1H), 2.39–2.31 (m, 1H), 1.54 (td, $J = 12.4$, $J = 2.4$ Hz, 1H); **^{13}C NMR** (126 MHz, $CDCl_3$) δ: 165.4, 160.6, 156.4, 146.8, 145.9, 132.0, 124.8, 111.0, 107.5, 107.4, 106.8, 105.4, 100.7, 78.5, 68.8, 61.1, 58.1, 57.3, 55.6, 55.5, 45.8, 30.2. **ESI-HRMS** *m/z* calculated for $C_{26}H_{28}NO_7^+$ $[M + H]^+$ 466.1860, found 466.1862.

Conversion of 1 nm and 2 nm to hydrochloride salts:

The synthesized derivatives 1 nm and 2 nm were dissolved in ethereal HCl and converted to the corresponding hydrochlorides.

ADME and drug-likeness prediction *in silico*

Evaluation of the drug-likeness and ADME features of novel compounds *in silico* is streamlining the progression of these compounds in the realm of drug discovery. Presently, the freely available SwissADME bioinformatics web tool (accessible at http://www.swissadme.ch/, Molecular Modelling Group of the Swiss Institute of Bioinformatics, Switzerland) was used to predict the physicochemical parameters and drug-likeness descriptors of compounds 1 nm and 2 nm. Their drug similarity was assessed based on Lipinski's rule of five, Abbott's bioavailability score (ABS), and SwissADME oral bioavailability criteria. Gastrointestinal (GI) absorption was predicted using the BOILED-Egg model, and the compounds were also evaluated for pan-assay interference structures (PAINS) as a medicinal chemistry filter, all conducted *in silico*.

Aqueous solubility evaluation of 1 nm and 2 nm *in vitro*

The water solubility of the two derivatives as their hydrochloride salts was determined following the method described by Pidany *et al*. (2024), with minor modifications (Pidany *et al*. 2024). Stock solutions of each test compounds were prepared at a concentration of 500 μM in ultrapure distilled water containing 5% dimethyl sulfoxide (DMSO). The maximum UV–Vis absorption wavelength was determined using a Shimadzu UV-1601 UV–Vis spectrophotometer (Shimadzu Corporation, Japan) and the measured wavelength was subsequently used as a reference to measure the absorbance across a range of concentrations. Seven serial dilutions were prepared from the stock solution with concentrations ranging from 250 to 1 μM. For each concentration, 2 ml was used to record the UV absorbance at the predetermined wavelength. The experiment was repeated as triplicate and the mean absorption was calculated for each concentration. The obtained data were used to generate a calibration curve by linear regression method. To determine water solubility, an oversaturated solution of each test compound was prepared and sonicated at 37°C for 10 min. The solution was then centrifuged at 8000 rpm for 10 min. The supernatant was collected for UV–Vis absorbance measurement. If the absorbance exceeded the range of the calibration curve, the solution was appropriately diluted. The maximum aqueous solubility concentration was calculated using the derived calibration formula.

Determination of antibacterial activity of 1 nm and 2 nm against selected reference and clinical isolate bacterial strains

The microdilution broth method was performed according to The European Committee on Antimicrobial Susceptibility Testing (EUCAST) instructions, with slight modifications (Hasselmann and Microbiology 2003). Eight reference bacterial strains purchased from the Czech Collection of Microorganisms (CCM, Czech Republic) or from the German Collection of Microorganisms and Cell Cultures (DSM, Germany), namely: *S. aureus* subsp. *aureus* CCM 4223 (ATCC 29213), MRSA CCM 4750 (ATCC 43300), *S. epidermidis* CCM 4418 (ATCC 12228), *Enterococcus faecalis* CCM 4224 (ATCC 29212), *E. coli* CCM 3954 (ATCC 25922), *K. pneumoniae* CCM 4415 (ATCC 10031), *Acinetobacter baumannii* DSM 30007, (ATCC 19606), and *P. aeruginosa* CCM 3955 (ATCC 27853). Clinical isolates were provided by the Department of Clinical Microbiology, University Hospital, Hradec Králové, Czech Republic. All mentioned strains were taxonomically classified by biochemical tests and MALDI-TOF (Microflex LH/SH MALDITOF, Bruker Biotyper 3.0 SW, Bruker Daltonic GmbH, Germany) instrumentation. The antibiograms of the strains were determined using the disc diffusion method following EUCAST recommendations (Matuschek *et al*. 2014).

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The cultivation of bacteria was performed in Cation-adjusted Mueller–Hinton broth (CAMHB, M-H 2 Broth, Sigma–Aldrich, USA) at 35 ± 2°C. The tested compounds were dissolved in DMSO (Merck, Germany) to produce stock solutions. The final concentration of DMSO in the testing medium did not exceed 1% (*v/v*) of the total solution composition and did not affect the growth of bacteria. Both positive (microbe solely) and negative (cultivation medium and DMSO) controls were included in the assays. Internal quality standard—ciprofloxacin (CIP, Merck, USA) was used in assays. Antibacterial activity expressed as MIC (reported in μM), was evaluated after 24 (48 h) of static incubation in a dark and humidified atmosphere at 35 ± 2°C. Visual inspection and spectrophotometric measurements (530 nm, Synergy HTX Multi-Mode Microplate reader, BioTek, USA) were used for MIC endpoint evaluation.

Determination of antibacterial activity of 1 nm and 2 nm against selected mycobacterial strains

The antimycobacterial assay was performed with rapidly growing *M. smegmatis* DSM 43465 (ATCC 607), *M. aurum* DSM 43999 (ATCC 23366) from German Collection of Microorganisms and Cell Cultures (DSM, Germany), and with an avirulent strain of *M. tuberculosis* H37Ra ITM M-006710 (ATCC 9431) from Belgian Co-ordinated Collections of Microorganisms (Belgium). The technique used for activity determination was the microdilution broth panel method using 96-well microtitration plates. The mycobacterial strains were cultured on Middlebrook 7H9 agar and suspensions were prepared in Middlebrook 7H9 broth (Merck, Germany) enriched with 0.4% glycerol (Merck, Germany) and 10% Middlebrook OADC growth supplement (Himedia, India). The final density of bacterial suspension was adjusted to 1.0 on the McFarland scale and diluted in the ratio of either 1:20 (for rapidly growing mycobacteria) or 1:10 (for slow growing mycobacteria) with broth.

The tested compounds were dissolved in DMSO (Merck, Germany), and then Middlebrook broth was added to obtain a concentration of 2000 μg ml^{-1}. Isoniazid (INH), Rifampicin (RIF), and CIP (Merck, Germany) were used as standards for the activity determination. The final concentration of DMSO did not exceed 2.5% (*v/v*) and did not affect the growth of all strains. Positive (broth, DMSO, bacteria) and negative (broth, DMSO) growth controls were included.

The antimycobacterial activity was expressed as MIC and was determined using Alamar Blue solution (Alamar Blue TM Cell Viability reagent, Thermo Fisher Scientific, USA). The value was read based on stain colour change (blue colour—active compound; pink colour—inactive compound). All experiments were conducted in duplicate.

Preliminary *in vitro* cytotoxicity or antiproliferative activity evaluation of compounds 1 nm and 2 nm

Compounds 1 nm and 2 nm were evaluated for preliminary off-target cytotoxic activity against a panel of nine selected human tumour and non-tumour cell lines. Jurkat (acute T-cell leukaemia), MOLT-4 (acute lymphoblastic leukaemia), A549 (lung adenocarcinoma), HT-29 (colorectal adenocarcinoma), PANC-1 (pancreatic epithelioid carcinoma), A2780 (ovarian carcinoma), HeLa (cervix adenocarcinoma), MCF-7 (breast adenocarcinoma), SAOS-2 (osteosarcoma), and MRC-5 (normal lung fibroblasts) were purchased from European Collection of Authenticated Cell Cultures (ECACC, UK) and cultured according to the guidelines from the provider. All cell lines were cultured at 37°C in a humidified incubator with 5% CO_2. For the experiments, cells were used in the maximum range of either 10 passages for the primary cell line (MRC-5) or 20 passages for the cancerous cell lines, in an exponential growth phase. Immediately prior to each experiment, the tested compounds and doxorubicin (used as a positive control) were freshly dissolved in DMSO (Sigma–Aldrich, USA) to prepare stock solutions at a concentration of 50 mM. These stock solutions were subsequently diluted in the appropriate cell culture medium to achieve the final working concentrations used in the assays. In this assay, the DMSO concentration was consistently maintained below 0.1%. Control cells were mock treated with DMSO (0.1%; control).

The WST-1 (Roche, Germany) reagent was employed to assess the cytostatic effect of the compounds using a 96-well-plate format (Sigma, USA). The cells were seeded at previously optimized densities (1×10^3 to 50×10^3 cells per well) and let to settle overnight. Following cell seeding, cells were treated with the respective compound derivative or doxorubicin stock solution, yielding final concentrations of 10 μM for the derivatives and 1 μM for doxorubicin. Treated cells were then incubated at 37°C in a humidified atmosphere containing 5% CO_2. At 48 h post-treatment, WST-1 reagent was prepared according to the manufacturer's protocol and added per well. Absorbance was measured at 440 nm on a Tecan Spark multimode microplate reader (Tecan Group, Switzerland). Experiments were run in triplicate and repeated independently at least three times. Results are presented as the proliferation percent relative to 0.1% DMSO treated control cells (set to 100%).

Descriptive statistics were calculated, and all charts were generated drawn in Microsoft Office 365 Excel (Microsoft, USA) or GraphPad Prism (GraphPad Prism 9.3.0, USA). The values are presented as mean ± standard deviation (SD) of triplicate measurements ($n = 3$), unless stated otherwise.

Evaluation of the IC_{50} values of 1 nm and 2 nm against HepG2 cells

Human hepatocellular carcinoma HepG2 cells (ATCC HB-8065; passage 20–25) were acquired from European Collection of Authenticated Cell Cultures (ECACC, UK) and were cultured in Minimum Essential Eagle Medium supplemented with 10% foetal bovine serum and 1% L-glutamine solution (Sigma–Aldrich, USA) at 37°C in a humidified incubator with 5% CO_2. For passage, the cells were treated with trypsin-ethylenediamine tetraacetic acid (EDTA, Sigma–Aldrich, USA) at 37°C and subsequently harvested. The treated cells with 1 nm and 2 nm were used as cytotoxicity screening group. The untreated HepG2 cells served as the control group, and a blank group was also included in assay. The cells were plated in a 96-well plate at a density of 10 000 cells per well and were incubated for 24 h. Afterwards, the cells were treated with tested compounds dissolved in DMSO with not exceeding

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concentration of 0.5% *v/v* for DMSO. The testing compounds were prepared at different concentrations in triplicates according to their solubility and incubated for 24 h in a humidified incubator with 5% CO_2 at 37°C. After incubation, a solution of thiazolyl blue tetrazolium bromide (Sigma–Aldrich, USA) in RPMI 1640 without phenol red (BioTech, Czech Republic) was added and incubated for 30 min under the previously described conditions. Subsequently, DMSO was used to dissolve the formed formazan crystals, and the absorbance of the samples was recorded at 570 nm (Synergy Neo2 Multi-Mode Reader NEO2SMALPHAB; Agilent BioTek, USA). The IC_{50} values were determined by nonlinear regression analysis of semi-logarithmic plots of compound concentration vs. absorbance percentage relative to untreated controls, using GraphPad Prism software (GraphPad Prism 9.3.0, USA). The results are expressed as the concentration of the compound required to reduce cell viability to 50% of the maximum observed viability (IC_{50}). The values are expressed as arithmetic means ± SD of triplicate measurements ($n = 3$).

Evaluation of the IC_{50} value of 1 nm against HK-2 cells

The cytotoxicity of compound 1 nm was assessed using the CellTiter 96 colorimetric assay (Promega, WI, USA). This assay is based on the reduction of tetrazolium dye MTS by viable cells to form a formazan product. The IC_{50} as the quantitative parameter was measured using the colorimetric assay. The IC_{50} value was calculated with GraphPad Prism software (GraphPad Prism 9.3.0, USA) using log(inhibitor) vs. normalized response model, which enabled the comparison of cytotoxic effect for the selected cell line.

The human kidney cortex proximal tubule HK-2 cell line (ATCC, USA) was cultured in Dulbecco's Modified Eagle Medium (DMEM) High Glucose, supplemented with 10% fetal bovine serum and 2 mM L-glutamine. (All cell culture consumables were purchased from Sigma–Aldrich, USA.) Cells were maintained at 37°C in a humidified incubator with 5% CO_2 and subcultured for experiments once they reached 80%–90% confluency.

For subculturing, the cells were harvested after trypsin-EDTA (Sigma–Aldrich, USA) treatment at 37°C. To evaluate cytotoxicity, the cells treated with the test compound served as experimental groups and the untreated HK-2 cells employed as controls. The cells were seeded with the density of 10 000 cells per well in a 96-well plate. The next day, the cells were treated with 1 nm dissolved in DMSO with the prepared concentrations of 0.0001, 1, 5, 10, 25, 50, and 100 μM in triplicates. Higher concentrations were excluded due to observed precipitation of 1 nm in the culture medium, starting at a concentration of 100 μM. Treated cells were incubated for the following 24 h. Control group representing 100% cell viability, 0% viability (cells treated with 10% DMSO), cell-free blanks, and vehicle controls were incubated in parallel. After 24 h of incubation in a humidified atmosphere containing 5% CO_2 at 37°C, the reagent from the kit CellTiter 96 AQueous One Solution Cell Proliferation Assay (CellTiter 96; Promega, USA) was added, followed by 2 h of incubation under the described condition. The absorbance of samples was then recorded at 490 nm in a spectrophotometer (Tecan, Infinite M200, Austria). The IC_{50} was determined by nonlinear regression analysis of a semi-logarithmic plot of incubation concentration vs. percentage of absorbance relative to untreated controls, using GraphPad Prism software (GraphPad Prism 9.3.0, USA).

Screening of *in vivo* toxicity of compound 1 nm using the invertebrate animal model *Galleria mellonella*

Galleria mellonella animals were reared in the laboratory of the Department of Biological and Medical Sciences, at the Faculty of Pharmacy in Hradec Králové, Charles University. The larvae were fed with an artificial diet according to Haydak (1936) and reared in the dark at 29°C. Only fully vital, cream-coloured larvae with body weight ranging from 300 to 320 mg were selected for the administration of test compounds before each experiment.

Briefly, the tested compounds were dissolved in DMSO and diluted to the required working concentration using a phosphate-buffered saline, pH 7.4 (Merck, USA). The final concentration of DMSO corresponded to 30% (*v/v*). The samples were administered into the larvae at a total volume of 10 μl per larva, into the haemocoel through the last left proleg using a Hamilton syringe. Two control groups (untreated control and buffer-injected control) were included in the *in vivo* toxicity assessment of compound 1 nm. The inoculated larvae and control groups were then incubated in Petri dishes at 37°C. The survival and conditions of larvae were recorded over a 120-h period (24, 48, 72, 96, and 120 h after administration). Death was defined as the complete loss of mobility, including a physical stimulus using a plastic pipette. The mortality for each dose/time interval was calculated, and survival Kaplan–Meier curves were designed *via* SW analysis (GraphPad Prism 9.3.0, USA).

Evaluation of the anti-staphylococcal activity of 1 nm in combination with selected commercially available antibacterial agents

The Checkerboard assay based on the microdilution method was employed as a screening approach to evaluate the antibacterial effect within the mutual interaction of two compounds with antimicrobial activity (1 nm vs. selected antibacterial drug). The assays were performed in Honeycomb plates (Oy Growth Curves, Finland) in a ten-by-ten well configuration. Six antibiotic agents used for the treatment of infections caused by Gram-positive bacteria CIP, vancomycin (VAN), LIN, RIF, trimethoprim-sulfamethoxazole (SXT), and tigecycline (TGC) were selected for combinations (all purchased from Sigma–Aldrich, USA). Within the checkerboard assays, the MRSA ATCC 43300 strain was used for detecting the antibacterial activity of the compounds in combination. The compounds were dissolved in DMSO (Sigma–Aldrich, USA) to produce stock solutions. The final concentration of DMSO in the testing medium did not exceed 1% (*v/v*) of the total solution composition and did not affect the growth of bacteria. The compounds were serially diluted in CAMHB in separate microtiter plates and then transferred to a Honeycomb plate at a 1:1 volume ratio. The 1 nm was transferred in a vertical direction and a selected antibacterial drug in a horizontal direction. Each well with a final volume of 200 μL compound mixtures was inoculated with the reference MRSA

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bacterial strain in CAMHB. The positive controls consisted solely of the test microbe, while the negative controls contained the cultivation medium only. The honeycomb plates were incubated in a Bioscreen C instrument (Oy Growth Curves, Finland) at 36.8 °C for 20 h. After the incubation, the percentage of growth inhibition was calculated in comparison to the positive controls (background values subtracted). Spectrophotometric data, optical density, was recorded at 580 nm. A visual inspection using a metabolic activity indicator, Alamar Blue (Alamar Blue TM Cell Viability reagent, Thermo Fisher Scientific, USA), was also employed for evaluating MIC endpoints. The total fractional inhibitory concentration index (FICI) was then used to interpret the checkerboard assays. The FICI was calculated according to the following formula: FICI = FIC(A) + FIC(B). FIC(A) and FIC(B) were calculated as follows: FIC (A or B) = MIC of drug A (or B) in combination/MIC of drug A (or B) alone. The results of the FICI were defined as follows: FICI $\leq$ 0.5, synergy; 0.5 $<$ FICI $\leq$ 1, additivity; 1 $<$ FICI $\leq$ 4, indifference; and FICI $>$ 4, antagonism.

Statistical analysis

Screening survival experiments with *in vivo* toxicity studies consisted of 84 individuals for evaluation of 1 nm and 2 nm ($n = 7$). All data outputs were analysed by GraphPad Prism (GraphPad Prism 9.3.0, USA). Data from survival experiments were subjected to the log-rank Mantel–Cox (curve comparison) test. Results were considered significant at a *P*-value $<$ 0.05 in all analyses.

Results and discussion

Primary scaffolds of different classes of AAs are constantly subjected to modifications to find new potential drugs. Development of narciclasine-4-*O*-β-D-glucopyranoside as a potential compound to combat brain tumours, semi-synthetic 11-*O*-(1-naphthoyl)ambelline and 11-*O*-(2-methylbenzoyl)ambelline with significant potency in butyrylcholinesterase (BuChE) inhibition (IC_{50} values $<$ 0.3 μM), and 3,11-dinicotyl-11-hydroxyvittatine with 16 times better IC_{50} than its basal alkaloid against *Plasmodium falciparum* F32 (IC_{50} value of 0.8 $\pm$ 0.06 μM) are successful examples of these alkaloids chemical modification (Ingrassia *et al*. 2009, Cedrón *et al*. 2012, Maríková *et al*. 2020). Additionally, antimicrobial property investigations have also been targeted in the structural optimization of AAs; in one study targeting columnaris disease in catfish caused by the Gram-negative bacterium *Flavobacterium columnare*, a carbamate analogue of lycorine exhibited promising activity against two isolated strains of this bacterium, i.e. MIC of 5.5 $\pm$ 0 mg l^{-1} for both (Tan et al. 2011). In another study aiming to identify novel natural-based antimycobacterial molecules, semi-synthetic esters of various classes of AAs were prepared, which in some cases resulted in markedly increased activity of the scaffold. Two compounds, 6-*O*-(4-butylbenzoyl)galanthamine and 6-*O*-(2-naphthoyl)galanthamine, with MIC values of 3.5 and 4.1 μM against *M. tuberculosis* H37Ra expressed the best antitubercular potencies. In the aforementioned study, seven semi-synthetic derivatives with montanine-type scaffold were included, among which ester derivatives with 4-*tert*-butylbenzoyl, 4-butylbenzoyl, 3,5-diethoxybenzoyl, and 1-naphthoyl on C2 position of 3-*O*-methylpancracine skeleton displayed MIC values $<$8 μM against *M. tuberculosis* H37Ra (Maafi *et al*. 2021a).

In this study, attention is given to two compounds, 1 nm and 2 nm, which are semi-synthetic derivatives featuring montanine-type scaffolds, bearing the characteristic 5,11-methanomorphantridine core. Their specific attributes, which determine their suitability as potential antimicrobial drug candidates, are thoroughly investigated in this work. Specifically, properties such as drug-likeness, cytotoxicity, toxicity *in vivo,* and antibacterial activity of 1 nm and 2 nm were studied.

Aqueous solubility and *in silico* drug-likeness prediction for 1 nm and 2 nm

Drug-like properties, derived from structures and characteristics of existing drugs and drug candidates, has been widely used to filter out undesirable compounds in the early phases of drug discovery. The initial concept of drug-likeness for orally active compounds is proposed by Lipinski, known as the Rule of Five, which defines four key physicochemical parameters: molecular weight $\leq$ 500, ClogP $\leq$ 5 (or MlogP $\leq$ 4.15), hydrogen bond donors $\leq$ 5, and hydrogen bond acceptors $\leq$ 10 (Lipinski *et al*. 2001, Lipinski 2004).

The evaluation of drug similarity and ADME attributes of 1 nm and 2 nm was performed *in silico* using the free online tool, SwissADME (Table 1) (Daina *et al*. 2017). None of the compounds showed any violations of Lipinski's and SwissADME oral bioavailability criteria, indicating the positive *in silico* assessment of these two as orally bioavailable (Lipinski *et al*. 2001, Daina *et al*. 2017). The ABS of 0.55 predicts that both compounds are likely to exhibit at least 10% of oral bioavailability in rats or possess identifiable permeability in Caco-2 cells, according to *in silico* assessment. Furthermore, the GI absorption according to BOILED-Egg method concurs with high absorption of 1 nm and 2 nm from the GI tract (Daina and Zoete 2016). These findings suggest the high bioavailability of both compounds and possibility of GI administration. PAINS are miscellaneous fragments prone to interfere with assay readouts, leading to false-positive results (Daina *et al*. 2017). Notably, no such fragments were detected in 1 nm and 2 nm.

Additionally, the aqueous solubility of both derivatives in their hydrochloride salt forms was determined using UV–Vis spectrophotometry. 1 nm exhibited slightly lower maximum solubility (4613.96 μM, equivalent to 2177.46 mg l^{-1}) in aqueous medium compared to 2 nm (4669.47 μM, equivalent to 2343.89 mg l^{-1}). These findings are further supported by the predicted log S (ESOL) values, which indicate a well-balanced lipophilic–hydrophilic character, highlighting their potential as promising drug candidates with moderate to sparing solubility. However, it should be noted that the measured solubility values may have been slightly overestimated in comparison to the true solubility in pure aqueous medium, due to the presence of DMSO in the testing medium.

Antibacterial potential of 1 nm and 2 nm

Total number of 32 bacterial strains were included in this study to determine the antibacterial potential of 1 nm and 2 nm (Table 2). In the study, 22 Gram-positive bacterial strains (4 reference representatives and 18 clinical isolates), 7 Gram-negative bacterial strains (4 reference strains and 3 clinical isolates), and 3 mycobacterial strains were included for determination of antibacterial activities.

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Table 1 Physicochemical, ADME, drug likeness, and medicinal chemistry characteristics of compounds 1 nm and 2 nm.

Parameters	1 nm	2 nm
MW	435.47 g/mol	465.50 g/mol
HBA	7	8
HBD	0	0
RB	5	6
TPSA	66.46 Å^2	75.69 Å^2
Log $_{Po/w}$ (XLOGP3)	2.78	2.75
Log $_{Po/w}$ (MLogP)	2.50	2.19
Log S (ESOL)	−4.24 (Moderately soluble)	−4.32 (Moderately soluble)
Csp^3	0.40	0.42
GI absorption	High	High
LP violation	Zero violation	Zero violation
OBR violation	Zero violation	Zero violation
ABS	0.55	0.55
PAINS	Zero alert	Zero alert

All parameters and descriptors are obtained from the SwissADME bioinformatics web tool.
MW, molecular weight; HBA, number of H-bond acceptors, HBD, number of H-bond donors, RB, number of rotatable bonds; TPSA, topological polarity surface area; Log Po/w (XLOGP3), logarithm of the octanol–water partition coefficient calculated using the XLOGP3 algorithm; Log Po/w (MLogP), logarithm of the octanol–water partition coefficient estimated using the MLOGP method; Log S (ESOL), aqueous solubility by the ESOL model; Fraction Csp3–Fraction of carbon atoms in sp3 hybridization; GI absorption, gastrointestinal absorption according to the white of the BOILED-Egg; LP Violation, number of the violations of Lipinski's rule of five (MW $\leq$ 500 g/mol, MlogP $\leq$ 4.15, HBD $\leq$ 5, HBA $\leq$ 10); OBR Violation, number of the violations of SwissADME oral bioavailability radar (lipophilicity: -0.7 $<$ XLOGP3 $<$ +5.0, 150 g/mol $<$ MW $<$ 500 g/mol, 20 Å^2 $<$ TPSA $<$ 130 Å^2, −6 $<$ Log S (ESOL) $<$ 0, 0.25 $<$ Fraction Csp3 $<$ 1, 0 $<$ RB $<$ 9); ABS, Abbott's bioavailability score; and PAINS, pan-assay interference compounds.

As shown in Table 2, the most potent antibacterial activity of 1 nm was registered against *S. aureus* (ATCC 29213, MIC_{1nm} ranged from 3.9 to 15.6 μM), MRSA (ATCC 43300; MIC_{1nm} ranged from 15.6 to 62.5 μM), and *S. epidermidis* (ATCC 12228; MIC_{1nm} ranged from 15.6 to 31.3).

The lowest MIC of 2 nm was registered against *S. aureus* (ATCC 29213) (MIC_{2nm} ranged from 7.8 to 31.3 μM), MRSA (ATCC 43300) (MIC_{2nm} ranged from 7.8 to 31.3 μM), and *S. epidermidis* (ATCC 12228) (MIC_{2nm} ranged from 15.6 to 31.3 μM). The promising antibacterial activity of candidate compounds 1 nm and 2 nm demonstrated against reference bacterial strains was corroborated by results obtained using bacterial strains isolated from clinical specimens. Susceptibility profiles of each bacterial isolate can be found in the Supporting Information (Supplement SA).

An anti-staphylococcal activity was confirmed in compound 1 nm, with MIC_{1nm} = 7.8 μM against *S. aureus* (141/16) and MIC_{1nm} = 15.6 μM against MRSA (153/16). Similarly, MIC_{2nm} = 15.6 μM was detected against *S. haemolyticus* (111/16) and MIC_{2nm} = 31.3 μM was revealed in two MRSA strains (153/16 and 154/16). Such results are in close accordance with the MICs of 1 nm and 2 nm against corresponding reference strains.

Interestingly, a promising activity revealed against *Enterococcus faecalis* (ATCC 29212, $MIC_{1nm)}$ = 31.3–62.5 μM MIC_{2nm} = 31.2 –125 μM,) was not confirmed in vancomycin-resistant bacterial isolates (2/21, 7/21, 8/21, 17/21, and 18/21). Accordingly, moderate activity against some Gram-negative reference strains, e.g. *K. pneumoniae* (ATCC 10031; MIC_{1nm} = 15.6–62.5 μM and MIC_{2nm} = 31.3–62.5 μM) was not detected against bacterial strains isolated from clinical specimens.

Moderate antimycobacterial effect in 1 nm and 2 nm was also revealed. The lowest MIC_{1nm} corresponding to 33.1 μM was revealed against *M. smegmatis*. MIC_{2nm} of 31.1 μM was registered against all three mycobacterial collection reference strains (*M. smegmatis*, *M. aurum*, and *M. tuberculosis* H37Ra.

In summary, in both 1 nm and 2 nm, a promising antibacterial activity was revealed against reference bacteria and bacterial strains isolated from clinical specimens. Furthermore, the antibacterial effect seems to be the most prominent specifically against staphylococci isolated from clinical specimens (MIC_{1nm} corresponded to 7.8–62.5 μM and MIC_{2nm} corresponded to 15.6–62.5 μM). Such results may initially appear as a less compelling, given that staphylococci are not always perceived as the most pressing threat. Nevertheless, these pathogens must continue to be considered, as they are characterized by evolving resistance mechanisms and remain clinically relevant. This is further underscored by the 2024 update of the WHO priority pathogens list, in which MRSA strains were again classified under the 'High Priority' category (WHO 2024).

Among all classes of AAs, montanine-type alkaloids are known as potent antimicrobial agents due to their reported activity against pathogenic bacteria. For example, Castilhos *et al*. (2007) studied the antibacterial potency of montanine against pathogenic *S. epidermidis*, (MIC = 15 μg ml^{-1}), *S. aureus* (MIC = 5 μg ml^{-1}), *E. coli* (MIC = 5 μg ml^{-1}), and *P. aeruginosa* (MIC = 20 μg ml^{-1}) (Castilhos *et al*. 2007). Despite these promising findings, this group remains largely unexplored with respect to its antimicrobial potential (Koutova *et al*. 2020).

Cytotoxicity profile determination of 1 nm and 2 nm

Montanine, the most well-known alkaloid of montanine-type AAs, is renowned for its pronounced antiproliferative effect against a range of cancer cell lines (Koutova *et al*. 2023). Given that our derivatives share the same core scaffold with this alkaloid, the preliminary cytotoxicity/anti-proliferative profiling was conducted against a panel of nine cancer cell lines and one non-cancerous cell line. The results from the screening are demonstrated in

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Table 2 Antibacterial 5activity of 1 nm and 2 nm against Gram-positive and Gram-negative bacterial reference strains, clinical bacterial isolates, and three mycobacterial reference strains.

	Origin	1 nm MIC (μM)	2 nm MIC (μM)	CIP MIC (μM)		
Gram-positive bacterial strains						
Staphylococcus aureus subsp. *aureus* (ATCC 29213)	Reference	3.9–15.6	7.8–31.3		1.5	
Methicillin-resistant *Staphylococcus aureus* subsp. *aureus*, MRSA (ATCC 43300)	Reference	15.6–62.5	7.8–31.3		0.8	
Staphylococcus epidermidis (ATCC 12228)	Reference	15.6–31.3	15.6–31.3		0.8	
Enterococcus faecalis (ATCC 29212)	Reference	31.3–62.5	31.3–125		3	
Staphylococcus capitis (ID No. 74/16)	Peripheral venous catheter	31.3	62.5		1.5–6	
Staphylococcus haemolyticus (ID No. 111/16)	Urine catheter	62.5	15.6		3–≥6	
Methicillin-resistant *Staphylococcus aureus* subsp. *aureus*, MRSA (ID No. 131/16)	Laryngeal swab	31.3	62.5		≥6	
Staphylococcus aureus (ID No. 136/16)	Laryngeal swab	62.5	62.5		0.8–1.5	
Methicillin-resistant *Staphylococcus aureus* subsp. *aureus*, MRSA (ID No. 138/16)	Sputum	62.5	62.5		>6	
Staphylococcus aureus (ID No. 141/16)	Eye swab	7.8	31.3–62.5		0.4–1.5	
Methicillin-resistant *Staphylococcus aureus* subsp. *aureus*, MRSA (ID No. 153/16)	Laryngeal swab	15.6	31.3		>6	
Methicillin-resistant *Staphylococcus aureus* subsp. *aureus*, MRSA (ID No. 154/16)	Bronchoalveolar lavage	31.3–62.5	31.3		>6	
Staphylococcus epidermidis (ID No. 196/16)	Drain	62.5	500		1.5–6	
Vancomycin-resistant *Enterococcus faecium* (ID No. 198/16)	Catheter tube	>500	>500		6	
Vancomycin-resistant *Staphylococcus aureus* (ID No. 206/16)	ND	62.5	62.5		0.8–3	
Staphylococcus hominis (ID No. 109/18)	Central venous catheter	62.5	62.5		3	
Vancomycin-resistant *Staphylococcus aureus* (ID No. 203/19; NIPH)	NIPH	62.5	62.5		0.8	
Vancomycin-resistant *Enterococcus faecalis,* VRE (ID No. 2/21)	ND	>250	>250		>6	
Vancomycin-resistant *Enterococcus faecalis,* VRE (ID No. 7/21)	ND	>250	>250		>6	
Vancomycin-resistant *Enterococcus faecalis,* VRE (ID No. 8/21)	ND	>250	>250		>6	
Vancomycin-resistant *Enterococcus faecium,* VRE (ID No. 17/21)	Urine	>250	>250		>6	
Vancomycin-resistant *Enterococcus faecium,* VRE (ID No. 18/21)	Urine	>250	>250		>6	
Gram-negative strains						
Escherichia coli (ATCC 25922)	Reference	>125	>125		1.5	
Klebsiella pneumoniae (ATCC 10031)	Reference	15.6–62.5	31.3–62.5		1.5	
Acinetobacter baumannii (ATCC 19606)	Reference	>125	>125		3	
Pseudomonas aeruginosa (ATCC 27853)	Reference	>125	>125		0.8	
Klebsiella pneumoniae (ID No. 13/21)	Urine	>500	>500		>6	
Klebsiella pneumoniae (ID No. 14/21)	Rectal swab	>500	>500		>6	
Klebsiella pneumoniae (ID No. 15/21)	Rectal swab	>500	>500		6	
Mycobacterial strains				INH MIC (μM)	RIF MIC (μM)	CIP MIC (μM)
Mycolicibacterium smegmatis (ATCC 607)	Reference	33.1	31.1	113.8	3.8	0.2
Mycolicibacterium aurum (ATCC 23366)	Reference	66.2	31.1	28.4	0.5	0.06
Mycobacterium tuberculosis H37Ra ITM-M006710 (ATCC 9431)	Reference	66.2	31.1	2.2	0.004	0.9

The antibacterial activity was determined by the microdilution broth method according to EUCAST recommendations. The MIC was evaluated by visual inspection *via* the metabolic indicator Alamar Blue, and by spectrophotometric measurement. The results were read after 24 h of incubation for bacterial strains and 48 h of incubation for *M. smegmatis*, 72 h for *M. aurum*, and 120 h for *M. tuberculosis* at 37°C.
ATCC, American Type Culture Collection; CIP, ciprofloxacin; ID No., internal laboratory identification number; INH, isoniazid; MIC, minimum inhibitory concentration; MRSA, methicillin-resistant *Staphylococcus aureus*; ND, not determined; NIPH, The National Institute of Public Health in Prague, Czech Republic; RIF, rifampicin; VR, vancomycin-resistant; and VRE, vancomycin-resistant *Enterococcus* sp.

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Table 3 Cell proliferation registered 48 h after exposition of human cell lines to 1 nm and 2 nm at final concentration corresponding to 10 μM.

	Cell types									
	Jurkat	MOLT-4	A549	HT-29	PANC-1	A2780	HeLa	MCF-7	SAOS-2	MRC-5
1 nm	124 ± 7%∗	85 ± 9%∗	100 ± 13%∗	77 ± 12%∗	110 ± 8%∗	87 ± 8%∗	104 ± 5%∗	88 ± 3%∗	103 ± 14%∗	98 ± 12%∗
2 nm	134 ± 12%∗	86 ± 5%∗	74 ± 20%∗	60 ± 16%∗	119 ± 16%∗	92 ± 22%∗	94 ± 8%∗	76 ± 11%∗	94 ± 17%∗	112 ± 7%∗
DOX	2 ± 0%∗	1 ± 1%∗	28 ± 26%∗	50 ± 2%∗	87 ± 5%∗	9 ± 4%∗	15 ± 2%∗	31 ± 7%∗	21 ± 3%∗	22 ± 3%∗

Values represent cell proliferation expressed as a percentage of the proliferation related to the untreated control cell line. Each value represents the mean of three independent experiments. Doxorubicin was used as a reference drug at final concentration corresponding to 1 μM.
%∗, percentage of the proliferation related to the untreated control cell line. DOX, doxorubicin; Jurkat, human acute T-cell leukaemia; MOLT-4, human acute lymphoblastic leukaemia; A549, human lung adenocarcinoma; HT-29, human colorectal adenocarcinoma; PANC-1, human pancreatic adenocarcinoma; A2780, human ovarian carcinoma; HeLa, human cervical adenocarcinoma; MCF-7, breast adenocarcinoma; SAOS-2, human osteosarcoma; and MRC-5, Human non-cancer fetal lung fibroblast cell line.

Table 4 Evaluation of *in vitro* IC_{50} values against HepG2 and HK-2 cells.

	IC_{50} (μM) against HepG2 cells	SI_{HepG2}*	IC_{50} (μM) against HK-2 cells	SI_{HK-2}*
1 nm	36.03 ± 5.17	9.2–0.6	>50	>12.8 to >0.8
2 nm	29.68 ± 6.95	3.8–0.5	n.d.	n.d.

Tetrazolium-based colorimetric assay was employed to assess IC_{50} values of 1 nm and 2 nm against HepG2, and 1 nm against HK-2 cell lines. The results are expressed as a mean of three replicate measurements. IC_{50}, half maximal inhibitory concentration; HepG2, human hepatocellular carcinoma cell line; and HK-2, normal human renal proximal tubular epithelial cell line. ∗Selectivity index (SI) is calculated as the IC_{50} against HepG2 cells or HK-2 cells in μM/MICs ≤ 62.5 in μM. n.d.: stands for 'not determined'.

Table 3. Treatment of cells with 10 μM concentrations of 1 nm or 2 nm did not result in significant antiproliferative activity. This outcome stands in contrast to our previous findings on the parent compound montanine, which was identified as the most potent isoquinoline alkaloid in that study, exhibiting notable cytotoxicity with IC_{50} values ranging from 1.04 to 2.30 μM. These results suggest that structural modification of montanine at the positions introduced in derivatives 1 nm and 2 nm may attenuate its cytotoxic activity (Koutova *et al*. 2023).

Hepatotoxicity and nephrotoxicity are critical markers to consider when evaluating candidate drugs during the development of novel antimicrobial agents. Although none of the derivatives exhibited any significant cytotoxicity at low doses *in vitro*, HepG2 cells were employed as a hepatotoxicity model to determine the IC_{50} values for both compounds (Table 4). 1 nm, with an IC_{50} of 36.03 ± 5.17 μM, displayed slightly lower toxicity towards HepG2 cells compared to 2 nm. Dose-response curves for compounds 1 nm and 2 nm against HepG2 cells are provided in Supporting Information (Supplement SB). Given its lower hepatotoxicity, 1 nm was further tested on the human kidney epithelial cell line HK-2.

Since in the higher concentrations (100 μM) of 1 nm the solubility of the compound in the culture medium limited the measurement, the IC_{50} value of 1 nm for HK-2 cells was determined to be greater than 50 μM (Fig. 2).

To discuss the *in vitro* hepatotoxicity and nephrotoxicity findings, the selectivity index (SI) is defined as the ratio of IC_{50} value against HepG2 cells or HK-2 cells to the MIC ≤ 62.5 μM, which serves as a useful *in vitro* indicator of a compound's therapeutic potential. An SI value >10 is generally indicative of a compound with both sufficient efficacy and an acceptable safety margin (Sanchez and Kouznetsov 2010). Our results suggest that 1 nm

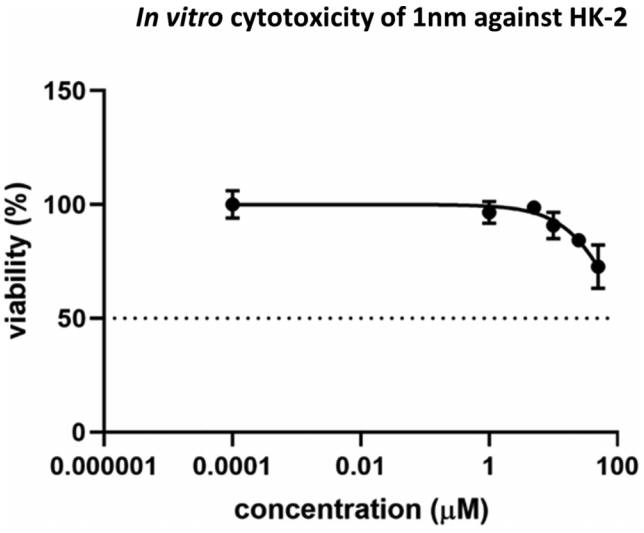


Figure 2 Evaluation of *in vitro* cytotoxicity of 1 nm against the HK-2 cell line. The cytotoxic effect was expressed as the IC_{50} value, i.e. the concentration of 1 nm that induced a cytotoxic effect in 50% of the cells. The IC_{50} parameter was calculated using GraphPad Prism 9.3.0 software (GraphPad Software, Inc., USA).

with SI_{HepG2} ranging from 9.2 to 0.6 has a better safety profile than 2 nm, while maintaining a selectivity range of >12.8 to >0.8 for HK-2 cells.

Given the promising antibacterial activity, better aqueous solubility and more favourable hepatotoxicity profile, it was ultimately decided that derivative 1 nm would be included in subsequent evaluations.

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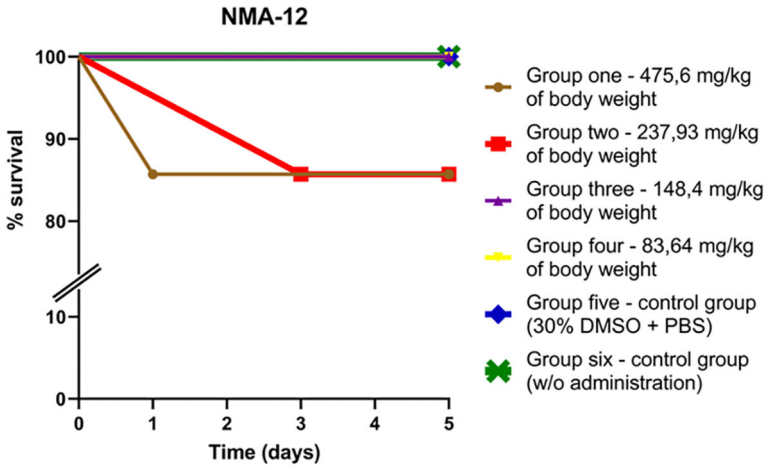


Figure 3 Survival curves of animal model, *Galleria mellonella*, after intra-haemocoel administration of 1 nm. After administration, larvae were incubated at 37°C for 5 days and inspected after 24, 48, 72, 96, and 120 h of incubation.

In vivo toxicity of 1 nm using *Galleria mellonella*

In vivo toxicity of 1 nm was evaluated using invertebrate animal model, *Galleria mellonella.* The larvae were divided into groups according to the final volume of administered compound per kg of the animal's body weight. The highest administered doses corresponded to 475.6 mg per kg of body weight; the lowest doses had a final concentration of 83.64 mg per kg of body weight of larvae. The tested compound, 1 nm, was administered into the haemocoel through the last left proleg by an injection, which mimics the intravenous route of administration in mammals and reflects systemic toxicity. As it is presented in Fig. 3 and Supplement SC (Supporting Information), the median lethal dose leading to the death of 50% of the animals (LD_{50}) was not reached after the intra-haemocoel administration of 1 nm in all groups. Therefore, it can be concluded that the LD_{50} of 1 nm within the 24- to 120-h post-exposure interval exceeds 475.6 mg per kg of larval body weight.

The *in vivo* toxicity data were also statistically analysed using the paired Log-rank (Mantel–Cox) test (Supplement SD in Supporting Information). Following the intra-haemocoel administration of 1 nm derivative, no statistically significant differences in survival were observed among any of the experimental groups.

Based on these findings, 1 nm could be categorized within the GHS (Globally Harmonized System) as class 4, which represents non/low toxic compounds (Ignasiak and Maxwell 2017).

Exploring the potential of the nm1 as a synergistic adjuvant in antimicrobial therapy

Synergistic interaction between two compounds, in which one agent enhances the effect of the other and they act more efficiently together than as individuals, inspired many scientists to examine the significance of synergistic acting of plant-derived compounds and traditional antibiotics. Synergistically active plant-based compounds usually represent structurally diverse biologically active secondary metabolites.

Many researchers have proven synergistic effects of plant-based compounds in combination with common antibiotics against Gram-positive and Gram-negative species (Stefanović 2018). For example, a combination of grape pomace extracts and β-lactams, quinolones, tetracyclines, and chloramphenicol acted in synergy against all *S. aureus* and *E. coli* strains (Sanhueza *et al*. 2017). Ethanol extract of *Punica granatum* rind possesses synergistic activity with CIP against resistant strain of *K. pneumoniae* (Rafiq *et al*. 2017).

Ideally, combination therapy lowers the effective concentration of the active drug compared to the concentration necessary for the compound to be efficient alone. Such mutually convenient relations between two compounds could also lead to potentially reduced toxicity of the employed drugs and diminished adverse effects. Furthermore, combination therapy can mitigate the risk of the development of AMR, especially in those compounds with different mechanisms of action.

Due to favourable biological properties and strong premise of a good anti-staphylococcal activity of 1 nm against some of the most worrisome Gram-positive pathogens with multidrug resistance profiles and limited antibiotic treatment options targeted on this bacterial group, our team suggests that the drug candidate could act successfully as an adjuvant compound with a beneficial mutual effect to commercially available antibiotics.

To determine the effect of 1 nm in combination with clinically approved antibacterial drugs, 6 different drugs with different mechanisms of action were selected. Namely, CIP, VAN, LIN, RIF, SXT, and TGC were included to reveal the possible promising mutual interactions.

Due to its continuous clinical relevance, MRSA (ATCC 433000) was selected for evaluation of the impact of mutual interaction of 1 nm and selected antibacterial drug(s), on antibacterial activity. According to FICI values, presented in Supplements SE–SJ (Supporting Information), concentration ratios of all employed combinations were evaluated as synergistic, additive, indifferent or antagonistic relation. Combination of CIP + 1 nm revealed additive effect in 6 concentration ratios ($MIC_{CIP+1\,nm} = 0.006/0.012/0.024/0.05/0.09/0.19 + 62.50\ \mu M$) and all other ratios were evaluated as indifferent. In the combination VAN + 1 nm, 5 concentration ratios were evaluated as additive

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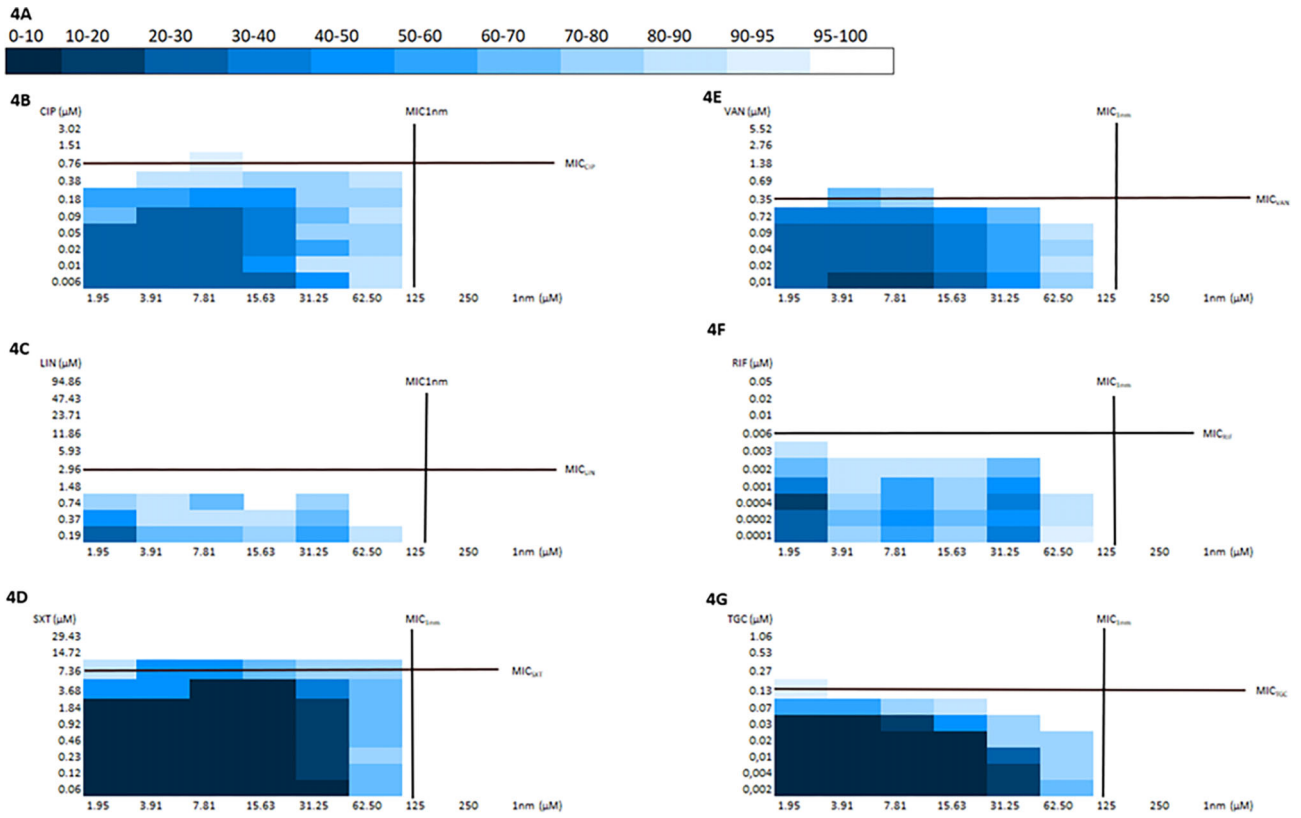


Figure 4 (a–g) Two-colour heat maps of checkerboard MIC assays. The percentage of growth inhibition (compared to positive growth control) of bacterial strain, methicillin resistant *Staphylococcus aureus*, MRSA (ATCC 43300), after 20 h of exposition at 37°C to 1 nm in combination with selected antibiotic drugs, expressed by colour spectrum. (a) Colour spectrum, (b) 1 nm in combination with ciprofloxacin (CIP), (c) vancomycin (VAN), (d) linezolid (LIN), (e) rifampicin (RIF), (f) trimethoprim-sulfamethoxazole (SXT), and (g) tigecycline (TGC). The black horizontal (selected antibacterial drug) and vertical (1 nm) lines denote the MIC of each compound alone.

($MIC_{VAN+1\,nm} = 0.01/0.02/0.04/0.09 + 62.50\ \mu M$ and $MIC_{VAN+1\,nm} = 0.17 + 31.25\ \mu M$), other concentration ratios showed indifferent effect. In the combination of LIN + 1 nm, synergistic effect was registered in one ratio ($MIC_{LIN+1\,nm} = 0.74 + 15.63\ \mu M$). Other concentration ratios in this combination were evaluated as additive. Combination of RIF + 1 nm revealed additive effect in all tested concentration ratios. Seven ratios with additive effect were revealed in the combination of SXT + 1 nm ($MIC_{SXT+MIC_{1\,nm}} = 0.06/0.12/0.23/0.46/0.92/1.84/3.67 + 62.50\ \mu M$), other ratios were evaluated as indifferent. And finally, 2 synergistic ratios ($MIC_{TGC+1\,nm}$ $0.02/0.03 + 31.25\ \mu M$) and 5 ratios with additive effect ($MIC_{TGC+1\,nm} = 0.002/0.004/0.008 + 31.25\ \mu M$, $MIC_{TGC+1\,nm} = 0.07 + 15.63\ \mu M$, and $MIC_{TGC+1\,nm} = 0.01 + 7.8\ \mu M$) were revealed in combination of TGC + 1 nm. Indifferent effect was assessed in all other concentration ratios of this combination.

The results of checkerboard assays are also presented in the form of heat maps (Fig. 4b–g). Each figure illustrates the percentual growth inhibition of the reference bacterial strain (MRSA) after being exposed to 1 nm in combination with selected antibiotic drugs. The percentage of growth inhibition is illustrated using a colour scheme (Fig. 4a), where the darkest shade represents the lowest inhibition and, therefore, the highest growth of the employed bacterial strain, while the lightest shade indicates the highest achieved inhibition and, consequently, the lowest growth of MRSA.

The heat maps clearly show that combination of 1 nm with antibiotics LIN (4C) or RIF (4F), causes a significant inhibition of the MRSA strain even at sub-inhibitory concentrations. These findings support the clinical relevance of these antibiotics. LIN is one of the last resort options for treating *S. aureus* infection, especially MRSA. Therefore, its use should be approached with caution and restraint, as the prevalence of *S. aureus* resistance to LIN remains low (Yang *et al*. 2025). In contrast, staphylococcal resistance to RIF is widespread. Therefore, RIF is commonly administered in combination with a second antibiotic drug, particularly in the treatment of staphylococcal biofilm-associated staphylococcal infections (Musumeci *et al*. 2025). The observed additive effect of 1 nm with both LIN and RIF suggests that 1 nm could potentially serve as a valuable adjuvant, enhancing the efficacy of existing antibiotics and helping to mitigate the development of AMR.

Conclusion

The prominent aim of this study was to investigate specific attributes of semi-synthetic montanine-type derivatives, labelled 1 nm and 2 nm, to evaluate their potential as adjuvant compounds for clinically used antibiotics. Our results demonstrate that 1 nm exhibits promising antibacterial activity, particularly against clinically relevant staphylococci, including MRSA strains. This narrow

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spectrum profile may help maintain microbiome balance. Both 1 nm and 2 nm displayed low cytotoxicity *in vitro* against a panel of cancerous and one non-cancer human cell line, while 1 nm showed more favourable cytotoxicity assessment against HepG2 cells with better selectivity, in comparison to 2 nm. Furthermore, the agreeable *in vitro* cytotoxicity assessment of 1 nm against HK-2 cells, and its low toxicity using the *Galleria mellonella* model, indicates favourable safety profile of 1 nm. Notably, this compound showed synergistic or additive effects when combined with several clinically important antibiotics, including LIN and RIF, suggesting its potential to enhance the efficacy of existing antimicrobial therapies. Though the *in silico* ADME parameters of both compounds suggest them as drug-like, the main limitation of these compounds may be their moderate solubility in aqueous media, which could affect the bioavailability. Overall, 1 nm represents a strong candidate for further development as an adjuvant antimicrobial agent, combining efficacy, safety, and synergistic potential with standard antibiotics, thus fulfilling a key requirement considered in the development and research of antimicrobial drugs.

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Author contributions

Adéla Diepoltová (Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing), Negar Maafi (Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing), Ondřej Janďourek (Investigation, Writing – review & editing), Katerina Hradiská Breiterová (Investigation, Writing – review & editing), Pavlína Vávrová (Investigation), Pavel Bárta (Investigation, Writing – review & editing), Darja Koutová (Investigation, Writing – review & editing), Radim Havelek (Investigation, Writing – review & editing), Marcela Vejsová (Investigation), Barbora Voxová (Investigation, Writing – review & editing), Petr Nachtigal (Writing – review & editing), Lucie Cahlíková (Conceptualization, Writing – review & editing), Klára Konečná (Conceptualization,Investigation,Writing – review & editing)

Supplementary material

Supplementary material is available at *Journal of Applied Microbiology* online.

Conflicts of interest

None declared.

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Data availability

The data underlying this article are available in the article and in its online Supplementary material.

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